Experiment 1: Electronic Components

1. **What is the difference between active and passive components?**
   Passive components give no power gain (resistor, capacitor, inductor). Active components can control voltage or current and amplify or switch (transistor, diode, IC).
2. **What is the function of a resistor?**
   It opposes current flow, limits current and drops voltage, dissipating energy as heat.
3. **How do you read a 4-band resistor?**
   The first two bands are digits, the third is the multiplier, and the fourth is the tolerance (gold ±5%, silver ±10%).
4. **What is the value of a resistor with Brown-Black-Orange-Gold bands?**
   $10 \times 10^3 = 10\ k\Omega \pm 5\%$.
5. **What does the SMD code 472 mean? What about 4R7 and 4702?**
   $472 = 47 \times 10^2 = 4.7\ k\Omega$. 4R7 = 4.7 Ω. $4702 = 470 \times 10^2 = 47\ k\Omega$.
6. **What is a capacitor, and what is the dielectric?**
   It stores energy in an electrostatic field between two plates. The dielectric is the insulating material between the plates.
7. **What does the code 104 on a ceramic capacitor mean?**
   10×10^4 pF = 100,000 pF = 100 nF = 0.1 µF.
8. **How do you identify the polarity of an electrolytic capacitor?**
   The negative lead is marked with a stripe, and the shorter lead is usually negative. It must be connected in the correct polarity.
9. **What is an inductor, and how are inductors classified?**
   A coil that opposes changes in current using its magnetic flux. By core type: air, iron, ferrite and powder core.
10. **On what principle does a transformer work?**
    Mutual induction. It transfers energy between circuits without changing frequency.
11. **What are the diode forward voltage drops for silicon, germanium and LED?**
    Silicon is about 0.6–0.7 V, germanium about 0.2–0.3 V, and LED about 1.8–3 V depending on colour.
12. **What is a Zener diode used for?**
    Voltage regulation. It works in reverse breakdown.
13. **What are the terminals of a BJT, and which types exist?**
    Emitter, base and collector. Types are NPN and PNP.
14. **How does a FET differ from a BJT?**
    A FET is voltage-controlled and unipolar with very high input resistance. A BJT is current-controlled and bipolar.
15. **What is a fuse, and how does it work?**
    A low-melting-point alloy strip connected in series. It melts and breaks the circuit when current is excessive.

16. **What is a relay?**
    An electrically operated switch that controls a high-power circuit using a low-power signal.
17. **What is the use of a heat sink?**
    It removes heat from power components to keep them cool.
18. **What is the use of a quartz crystal?**
    It acts as a precise resonator for oscillators and clock circuits.

Experiment 2: Symbols, EDA Tools and Datasheets

19. **Which standards define electronic symbols?**
    IEEE Std 315 and the British Standard BS 3939. BIS is the Indian standard.
20. **How do you show joined and unjoined wires in a circuit diagram?**
    Joined wires have a dot at the junction. Unjoined wires cross without a dot, or with a small hop.
21. **Draw the symbols for a resistor, capacitor, diode, LED and Zener diode.**
    Practise these from pages 12–13 of your manual.
22. **What is the difference between a rheostat and a potentiometer?**
    A rheostat is a two-terminal variable resistor for controlling current. A potentiometer is a three-terminal device used as a voltage divider.
23. **What is EDA?**
    Electronic Design Automation: software tools for designing circuits, PCBs and ICs.
24. **Which EDA tool did you use?**
    Proteus 8 Professional, for schematic capture and simulation.
25. **What is a datasheet, and why is it important?**
    A manufacturer's document giving ratings and characteristics. It is used to select the right component and operate it safely.
26. **What is the difference between maximum ratings and electrical characteristics?**
    Maximum ratings are limits that must never be exceeded. Electrical characteristics are the normal operating values.
27. **From the 1N4001–1N4007 datasheet, what is the peak reverse voltage of 1N4007? What is the average forward current?**
    PIV is 1000 V and Io is 1 A.
28. **How do you identify the cathode of a diode?**
    By the band marked on the body.
29. **What is the VCEO of BC107?**
    45 V.
30. **What is the hFE of a transistor?**
    The DC current gain: collector current divided by base current.
31. **What is the power dissipation rating of a 1N746 Zener?**
    500 mW.

32. **What type of device is the 2N2646, and where is it used?**
    A UJT (unijunction transistor), used in oscillators and thyristor trigger circuits.
33. **What is the supply voltage range of the 555 timer and the µA741?**
    The 555 runs up to a maximum of 16 V. The µA741C has a maximum supply of ±18 V.

Experiment 3: Instruments and Tools

34. **What is a CRO, and what does it display?**
    A cathode ray oscilloscope displays voltage on the Y-axis against time on the X-axis.
35. **What are the main parts of a CRT?**
    Electron gun, deflection plates (X and Y), and a phosphor screen.
36. **What does the Time/Div knob do? What does Volts/Div do?**
    Time/Div sets the horizontal time scale. Volts/Div sets the vertical voltage scale.
37. **What is the purpose of the calibration point?**
    It gives a standard square wave to check the probe and scope calibration.
38. **Why is a 10× probe used?**
    It reduces the signal to 1/10, reduces circuit loading and suits high-frequency signals.
39. **What is the function of the trigger?**
    It stabilises the waveform by starting the sweep at the same point of the signal.
40. **How can you measure frequency and peak-to-peak voltage using a CRO?**
    Frequency = 1 / (horizontal divisions × Time/Div). Vpp = vertical divisions × Volts/Div.
41. **What is a function generator?**
    An instrument that produces sine, square and triangular waves over a range of frequencies.
42. **What is the frequency range of the function generator you used?**
    1 Hz to 2 MHz for the APLAB 2011A. The Scientech generator goes up to 10 MHz for sine waves.
43. **What is DC offset?**
    A DC level added to the output signal for biasing.
44. **What does a multimeter measure?**
    Voltage, current and resistance, and also continuity, diode junctions and hFE.
45. **What is meant by a 3½ digit display?**
    Three full digits and one half digit that can only show 0 or 1, giving a maximum reading of 1999.
46. **Why should the iron tip be cleaned and tinned?**
    For good heat transfer and to avoid impurities in the joint.
47. **What is the difference between a soldering iron and a soldering gun?**
    An iron is pencil-shaped for precise work. A gun has a larger, high-wattage tip heated by direct current flow.
48. **What is the use of a desoldering station?**
    It removes solder from joints using heat and a vacuum pump.

49. **What are wire strippers, pliers, tweezers and crimpers used for?**
    Stripping insulation, holding and bending wires, picking up small components, and joining a wire to a connector by deformation.

Experiment 4: Testing of Components

50. **How do you measure voltage with a multimeter?**
    Connect it in parallel across the component, with the red lead in V-Ω and the black lead in COM.
51. **How do you measure current?**
    Connect the meter in series with the load, after opening the circuit.
52. **Why should you measure resistance only with the circuit power off?**
    Power in the circuit gives wrong readings and can damage the meter.
53. **How do you test a diode with a multimeter?**
    Set the meter to the diode position. A forward reading shows the drop (about 0.6 V for silicon). It reads over-range in reverse. If it reads over-range both ways, the diode is open. If it reads low both ways, it is shorted.
54. **How do you test a BJT?**
    Check the base-emitter and base-collector junctions like diodes, and check collector-emitter for a short.
55. **How do you identify the terminals of a transistor?**
    Use the datasheet pinout, or the multimeter test and the hFE socket.
56. **What does the continuity test indicate?**
    A beep when the resistance between the probe tips is low (below about 100 Ω).
57. **Why should you discharge capacitors before testing?**
    Stored charge can damage the meter or give a shock.

Experiment 8: Assembling Circuits

58. **What is an astable multivibrator?**
    A circuit with no stable state that switches continuously between two states, producing a square wave.
59. **What are the three modes of a 555 timer?**
    Astable, monostable and bistable.
60. **Which components fix the frequency of an astable 555 circuit?**
    Resistors R1 and R2, and capacitor C.
61. **What are the parts of a regulated power supply?**
    Transformer, rectifier, filter capacitor and regulator (Zener or IC).
62. **What is the function of the filter capacitor?**
    It smooths the pulsating DC from the rectifier.
63. **What is the purpose of the coupling capacitor in an RC-coupled amplifier?**
    It blocks DC and passes the AC signal between stages.

64. **What is the purpose of the emitter bypass capacitor?**
    It prevents loss of gain by bypassing AC signals around the emitter resistor.
65. **Write the truth table of AND and NAND gates.**
    AND outputs 1 only when both inputs are 1. NAND is the inverse of AND